Answer **all** the questions in this section in the spaces provided.

1 **(a)** Explain why the unit of pressure is a derived unit.

.....

.....

.....[2]

(b) A physical equation is of the form

$$Q = \frac{kr^4(p_1 - p_2)}{L}$$

where k is a dimensionless constant,

$r = (2.07 \pm 0.04)$ mm,

$p_1 = (127 \pm 2)$ kPa,

$p_2 = (105 \pm 1)$ kPa,

$L = (200 \pm 8)$ mm.

Calculate the fractional uncertainty in Q.

fractional uncertainty in Q = ... [3]

[Total: 5]